2026

IRoC-U

ISRO Robotics Challenge - URSC

Let's build a space robot

Proposal Report

For IRoC-U 2026

by

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1. Description of ASCEND

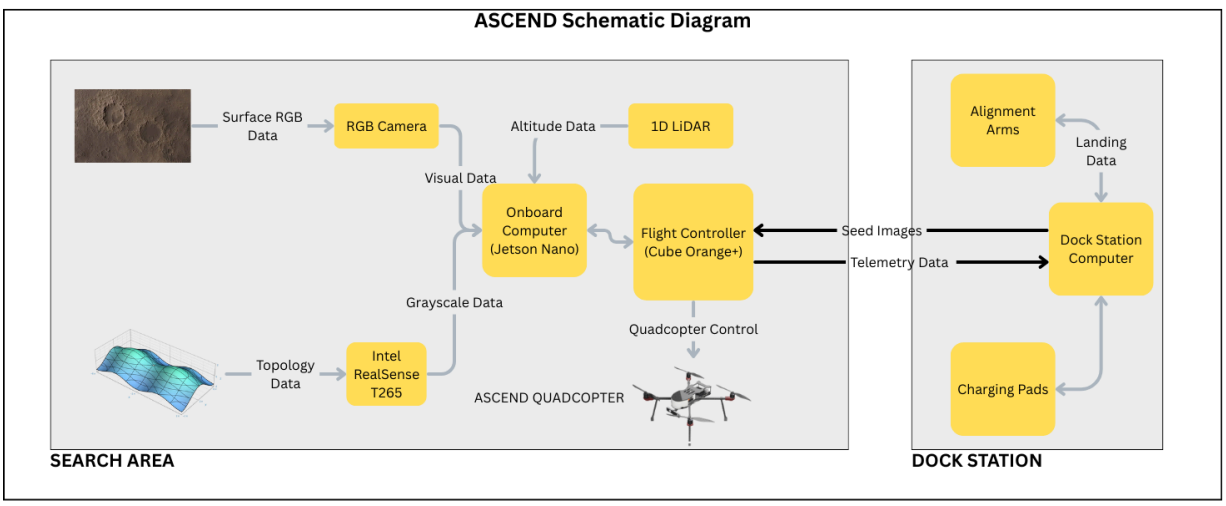

Fig 1.1: Schematic Diagram

ASCEND requires an autonomous micro-UAV to operate in a Mars-like, GPS-denied environment and perform terrain exploration, target detection, and precision return-to-base without human intervention. The UAV must rely entirely on onboard sensing, state estimation, and autonomous control to complete the mission.

Fig 1.2: Front View

Fig 1.3: Top View

Fig 1.4: Isometric View	Fig 1.5: Side View

Team Challengers proposes a fully autonomous quadcopter–docking system designed specifically for GNSS, marker, and external-aid-denied operation. The system consists of an X-configuration quadcopter optimized for precision flight and an autonomous docking station that enables post-mission data transfer and battery recharging.

System Overview

- X-configuration quadcopter capable of stable hover and low-speed maneuvering
- Fully onboard autonomy for takeoff, exploration, target detection, return, and landing
- Vision- and inertial-based navigation without reliance on GPS or external infrastructure
- Autonomous docking station for alignment, charging, and data exchange

Platform Selection and Feasibility

The proposed UAV platform was evaluated against both commercial and academic quadcopter systems to assess feasibility, cost-efficiency, and mission suitability. As illustrated in **Fig. 1.6**, unlike high-end commercial platforms such as the DJI Matrice 100 or Parrot Anafi USA, the proposed system prioritizes modularity, affordability, and full autonomy. This makes it particularly suitable for experimental research, iterative development, and competition environments where flexibility and customization are essential.

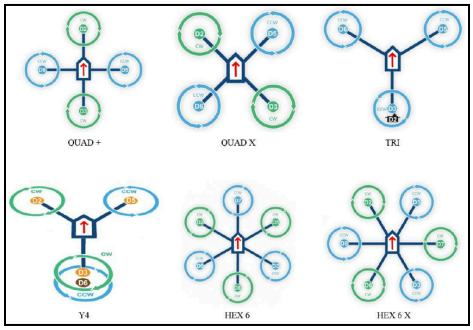

Fig 1.6

A quadcopter configuration was selected after evaluating multiple aerial platforms:

- Fixed-wing aircraft were unsuitable due to their inability to hover and perform precision landings

- Larger multirotor configurations increase redundancy but significantly raise mass, power consumption, and system complexity.
- The quadcopter provides the optimal balance between payload capacity, control accuracy, energy efficiency, and mechanical simplicity.

Payload and Design Constraints

The quadcopter is capable of carrying a payload of up to 500 g, sufficient for the onboard flight computer, vision sensors, and supporting electronics with adequate design margin. The complete system operates within the 2 kg mass limit and is constructed using off-the-shelf components, including high-torque BLDC motors, a carbon-fiber airframe, a Li-Po battery, and a Pixhawk Cube Orange flight controller. This approach balances durability, payload capability, and system reliability while simplifying integration and maintenance.

Docking Concept

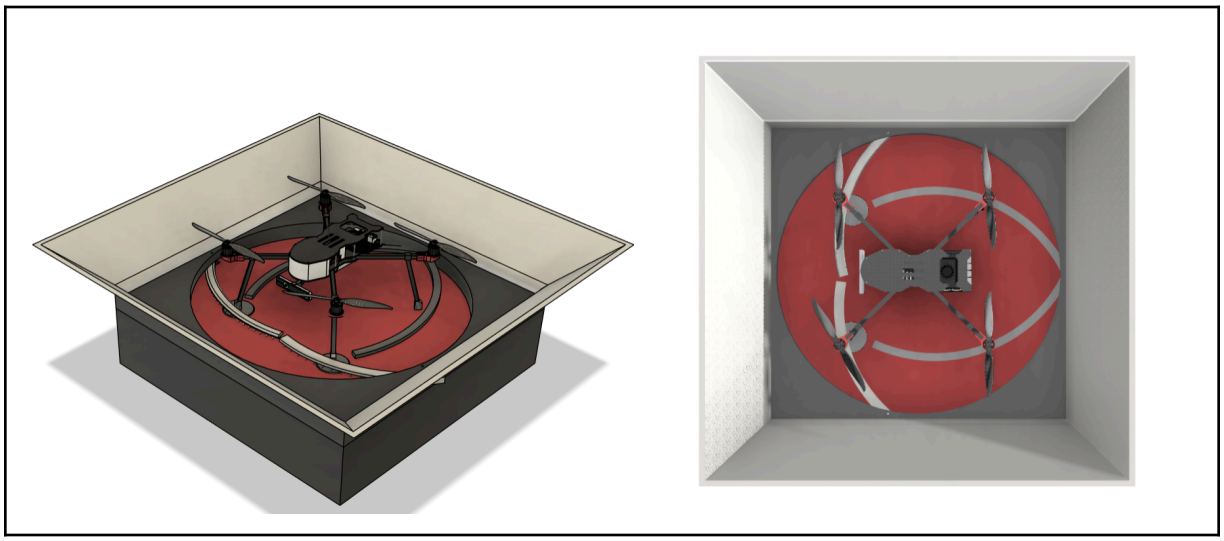

Fig 1.7

The docking station is designed to enable accurate, reliable, and fully autonomous landing and alignment even in GPS-, GNSS-, LPS-, and external-marker-denied conditions. As shown in **Fig. 1.7**, the system features a single circular base platform that

provides a symmetric and tolerance-friendly landing surface. Embedded circular metal contact plates act as the electrical charging interface, while the circular geometry minimizes sensitivity to yaw misalignment during descent. The docking station supports safe alignment, efficient power transfer, and seamless data exchange without human intervention.

Overall, the proposed ASCEND system emphasizes robust autonomy, precise control, and competition compliance, ensuring dependable mission execution in fully GNSS-denied environments.

2. System Architecture

The system architecture of ASCEND combines onboard sensing, high-level autonomy, and low-level flight control to enable reliable autonomous operation without external navigation aids. The architecture emphasizes modularity, real-time performance, and seamless sensor–controller integration.

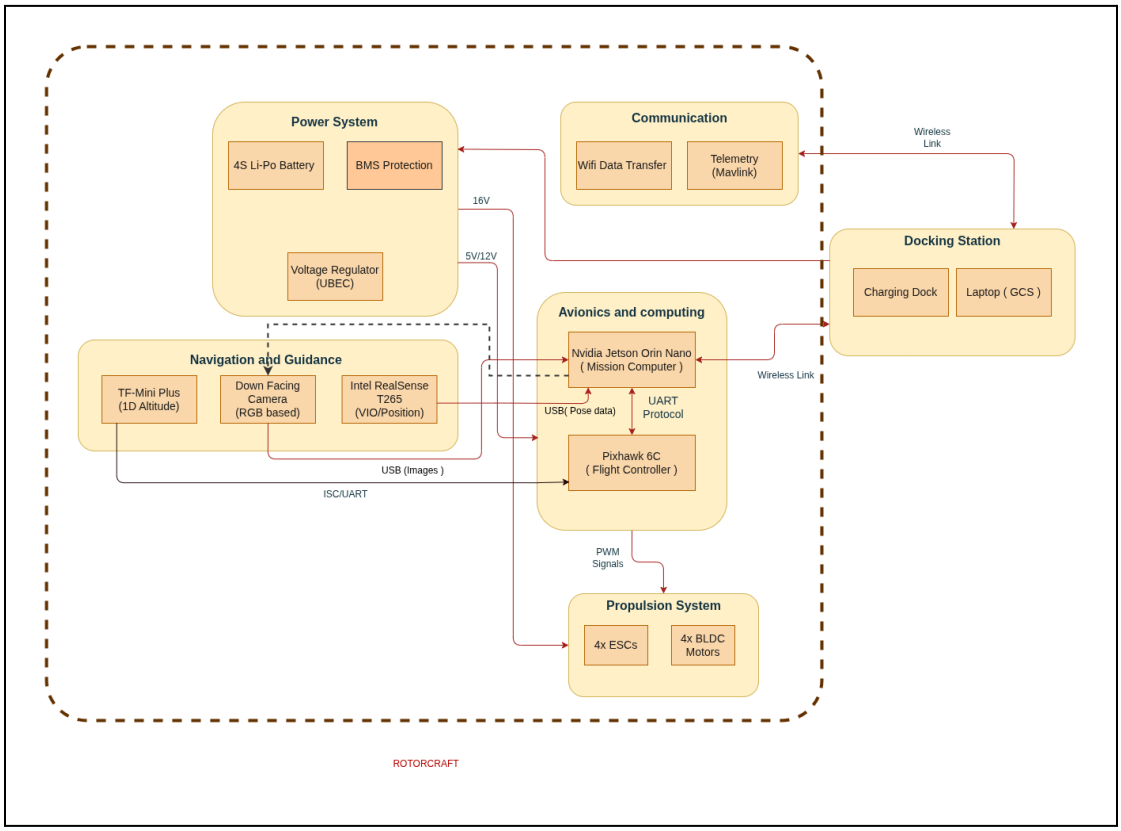

Figure 2.1: System architecture

The hardware design is centered around a dual-computer avionics stack, utilizing an Nvidia Jetson Orin Nano as the primary mission computer and a Pixhawk 6C for real-time flight control. Power management is handled by a 4S Li-Po battery system equipped with BMS protection, which distributes raw voltage to the propulsion system's ESCs and motors while routing regulated 5V and 12V lines to the sensitive electronics. The navigation suite integrates multiple sensor modalities, including a TF-Mini Plus for 1D altitude precision, a down-facing RGB camera for visual data, and the Intel T265 for VIO-based positioning, all interfacing with the mission computer via USB or UART. Communication is maintained through dual links: a telemetry module handles MAVLink traffic for flight data, while a high-bandwidth WiFi link transfers data between the onboard computer and the docking station's Ground Control Station (GCS).

Software Architecture (ROS 2 Implementation)

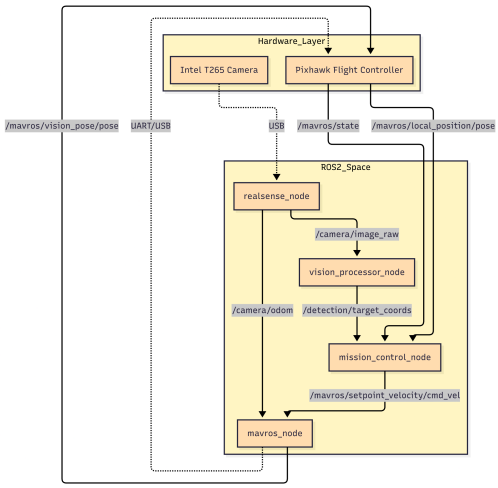

Figure 2.2

The software stack is built on a ROS 2 framework designed to decouple low-level flight stabilization from high-level perception and control logic. The architecture bridges the hardware layer—consisting of the Intel T265 and Pixhawk flight controller—with the application layer through a series of modular nodes. The realsense_node acts as the primary driver, ingesting raw data via USB and splitting it into two parallel streams: visual odometry (/camera/odom) for state estimation and raw video frames (/camera/image_raw) for the perception pipeline. While the mavros_node handles the critical MAVLink bridge to keep the Pixhawk updated with position data, the vision_processor_node executes object detection algorithms to output target coordinates. The central decision-making unit, the mission_control_node, subscribes to both the target coordinates and the drone's local position to generate real-time velocity commands (cmd_vel), closing the loop by sending these setpoints back to the flight controller for execution.

3. Identification of components with their specifications

3.1 Components Rotarcraft:

1. Mechanical Components

Frame: The ASCEND rotarcraft drone is built around a 500mm carbon fiber quadcopter frame, which forms the structural backbone of the vehicle. This custom quadcopter frame is fabricated from high-quality carbon fiber, making it lightweight, strong, and highly rigid. The carbon-fiber arms provide excellent structural integrity with minimal weight, ensuring improved flight efficiency and reduced vibrations. The optimized arm thickness and reinforced motor mounts enhance durability and minimize damage during hard landings. The rigid frame design improves overall stability, enabling precise control, faster forward flight, and better payload handling compared to conventional frames.

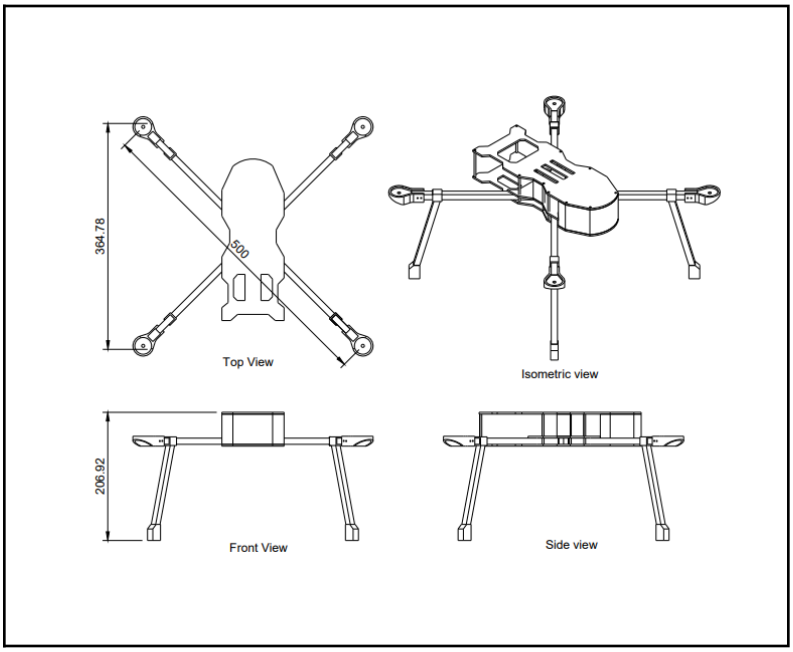

Fig 3.1.1

The drone's propulsion system consists of four AIR2216II KV920 brushless motors, each paired with an AIR 20A electronic speed controller (ESC). As illustrated in **Fig. 3.1.1**, these motors deliver sufficient thrust to support the drone and its onboard electronics while enabling precise and responsive maneuvering. The T1045 propellers (10×4.5) are arranged in a CW/CCW configuration to efficiently convert motor torque into lift. Although carbon fiber propellers were evaluated for their marginal efficiency gains, nylon-glass propellers were ultimately selected for their durability and ability to withstand minor impacts during testing and field operations.

2. Electrical Components

(a) Power Distribution

A 4S 3300 mAh Li-Po battery powers the entire drone, supplying all motors and ESCs. A separate 5V BEC (2–3A) provides clean, regulated power for the Jetson Orin Nano, the Cube Orange+ flight controller, and sensors. This setup ensures stable voltage and protects electronics from motor noise.

(b) Propulsion System

The drone uses four AIR2216II KV920 motors, drawing 8–12A in hover and up to 22A at full load. Each motor is controlled by a 20A ESC, capable of short 30A bursts. T1045 propellers generate the necessary lift and influence efficiency, motor load, and stability.

(c) Computation Hardware

The Cube Orange+ flight controller handles stabilization and uses about 250–300 mA at 5 V. The Jetson Orin Nano runs high-level processing and AI tasks, drawing 2–3 A at 5 V depending on workload.

3. Sensors.

(a) Intel RealSense T265 – Visual-Inertial Odometry (VIO) (Fig 3.1.2) Used for GPS-denied navigation. It estimates the drone's position and orientation using stereo vision + internal inertial sensors. Helps the drone know where it is in indoor/complex environments.	**(b) TFmini-S 1D LiDAR – Altitude Measurement (Fig-3.1.3)** Provides very accurate altitude at low heights (0–5 m), where barometers are unreliable. Essential for stable hovering and smooth autonomous landing.	**(c) Raspberry Pi Camera Module 3 – Vision & Detection (Fig 3.1.4)** Used for object detection, feature tracking, and target identification. Works with Jetson's computer vision models to locate the docking station or mission targets.

Position update: P(t)=P(t−1)+v(t)Δt+21a(t)Δt2 **Velocity update:** v(t) = v(t-1) + a(t)*Δt	**Time-of-flight distance:** d = (c * t) / 2 where d = distance c = speed of light t = round-trip travel time	**Harris corner response:** $R = det(M) - k(trace(M$ where M = structure tensor k ≈ 0.04

Appendix: Figures

 Fig 3.1.2:Intel RealSense T265	 Fig 3.1.3: TFmini-S 1D LiDAR	 Fig 3.1.4: Raspberry Pi Camera Module 3

Algorithms and Software

The software stack integrates low-level control, high-level perception, and autonomous mission management. Low-level flight stabilization is handled by the PX4 autopilot firmware on the flight controller, which manages motor commands and ensures stable flight. High-level autonomy runs on the Jetson Orin Nano using ROS 2, with custom nodes in C++ and Python handling SLAM, path planning, and feature detection.

Sensor data from the IMU, LiDAR, and T265 camera are fused using an Extended Kalman Filter (EKF) to generate an accurate estimate of the drone's position and orientation. Path planning algorithms, including A* and RRT, allow the drone to explore a designated area efficiently while avoiding obstacles. The software also handles autonomous target detection, coordinate logging, and image capture for validation at the docking station. For returning and landing, the LiDAR measures height, and the T265 camera assists with precise alignment, ensuring the drone docks safely. Overall, the algorithms are designed to allow the drone to operate fully autonomously, performing multiple sorties without human intervention.

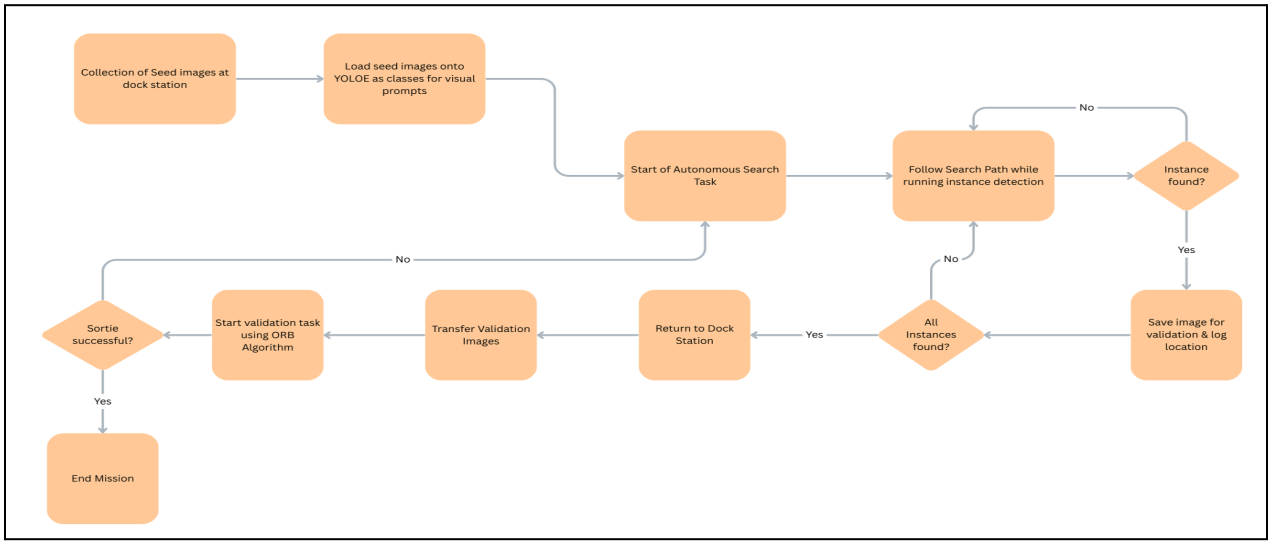

Fig 3.1.5

The mission workflow begins with the Seeding Task, where 3–5 sample feature images are collected at the dock station and loaded onto the YOLO-E model as visual prompts to define target classes. Once the Autonomous Search Task starts, the UAV follows a pre-defined path while running real-time instance detection to locate targets that are similar to the seed images. When an instance is found, the system saves a verification image and logs the local estimated coordinates; if not all instances are found, the UAV continues its search. After identifying the required targets, the UAV executes the Return Task to autonomously land at the dock station. The captured data is then transferred to the base station for the Validation Task, where an ORB pattern-matching algorithm compares the findings against the original seeds. If the sortie succeeds, the mission ends; if not, it is repeated.

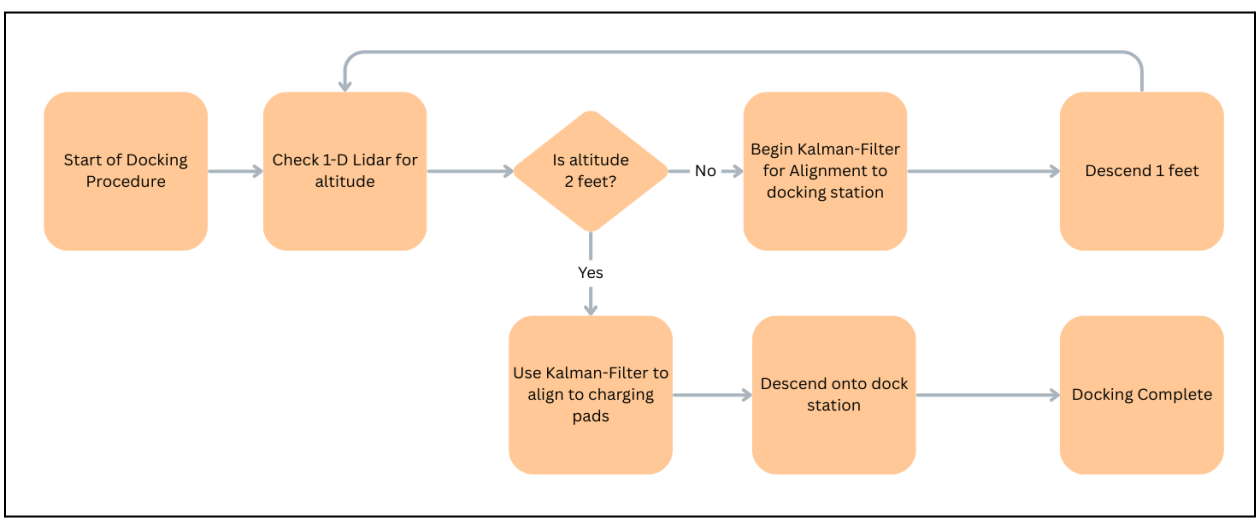

Fig 3.1.6

The process begins with the UAV using a 1-D Lidar to verify its altitude, complying with the prohibition on GNSS or external navigation aids. If the altitude is above 2 feet, a Kalman Filter aligns the UAV with the docking station while it descends in steps; once at 2 feet, the filter aligns it precisely with the charging pads for landing. This fully autonomous descent is essential, as manual alignment or human intervention is not allowed. After docking, the UAV transfers image data and coordinates to the base station while recharging through the designated interfaces.

3.2 Components Dockstation:

Cleaning and Locking Mechanism

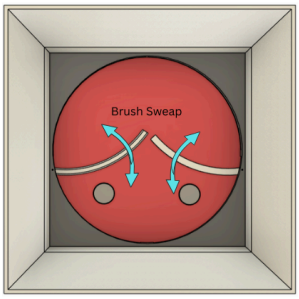

Fig 3.2.1: Cleaning

Fig 3.2.2: Aligning

Before landing, the docking station cleans the drone's charging contact surfaces to ensure a reliable electrical connection. The ESP32 controller receives a landing-approach command from the docking station computer, after which the cleaning mechanism is activated. After landing, a locking and alignment mechanism holds the drone firmly in place and corrects minor positional or orientation misalignment.

Charging System

Once proper alignment is achieved, the charging system establishes electrical contact and safely charges the drone's battery while it remains docked. The drone's landing legs are fitted with metallic contact pads, which interface with the docking station's charging contacts to ensure a reliable and stable electrical connection during charging.

- Battery energy capacity: 48.84 Wh (14.8 V, 3.3 Ah)
- Charging current: 6.0 A
- Contact resistance: 0.025 Ω

- Total charging time: ~39 minutes

Data Transfer

After docking, the drone connects to the docking station via wireless communication, enabling reliable data transfer between the aerial platform and the ground system. The data transfer system uses a combination of Wi-Fi and Bluetooth communication to ensure reliable, secure, and repeatable data exchange between the drone and the docking station after successful docking.

4. ASCEND Realization Plan

No.	Hardware Details	Procurement Source (Market/Fabrication/3D Printing….)	Specifications/ Realization Plan	Quantity
1	Frame	Fabrication (Carbon Fiber PLA)	450mm quadcopter frame, lightweight and rigid, includes mounting points for landing gear and electronics	1
2	Motors	Market	AIR2216II KV920 brushless, 4S LiPo compatible, 64g each, provides sufficient thrust for UAV	4
3	ESCs	Market	AIR 20A ESC, continuous current 20A, peak 30A, 53x22x7 mm, connects to motors via 3.5mm banana plugs	4
4	Propellers	Market	T1045 (10×4.5), nylon + glass fiber, 68g total for 4, CW/CCW setup	4
5	Battery	Market	4S 3300mAh LiPo, 16.8V full charge, provides power to motors, flight controller, and Jetson	1
6	Flight Controller	Market	Cube Orange flight controller module	1

7	Companion Computer	Market	Jetson Orin Nano, GPU-accelerated, runs ROS2 for SLAM, navigation, and mission control	1
8	Cooling Fan	Market	Fan for Jetson Orin Nano, maintains operating temperature during high-load computation	1
9	Visual Sensor	Market	Intel RealSense T265 camera provides visual-inertial odometry (VIO) for GPS-denied navigation	1
10	1D Lidar	Market	TFmini-S, for altitude measurement and safe landing	1
11	High-Current Switch	Market	XT60, 15–20A, for safe power on/off	1
12	Power Regulation / Power Electronics Component	Market	Holybro UBEC 12A: 3–14S input, selectable 6/7.2/8/9.2 V output, 12 A continuous (24 A peak), high-efficiency UBEC with full protection features.	1
13	Landing Gear with Metal contacts	Fabrication	Lightweight landing assembly to support charging after landing in the Docking station	1

5. Testing Plan

To ensure that the ASCEND system meets the requirements of autonomous navigation, perception, and mission execution without external navigation aids, a structured testing approach is adopted. Validation is carried out at subsystem and integrated system levels using high-fidelity PX4 SITL, ROS 2, and Gazebo simulations to verify sensing, autonomy, and safety prior to hardware deployment.

Subsystem-Level Testing

Camera (Downward-Facing RGB Camera) 1. Accuracy of visual feature detection 2. Consistency of image capture for documentation 3. Robustness under varying lighting and viewpoints	**Mapping and Perception Module** 1. Flat surface detection for landing 2. Obstacle detection and avoidance 3. Consistency of real-time map generation
IMU (Inertial Measurement Unit) 1. Stability under static and dynamic conditions 2. Accuracy of acceleration and angular rate measurements	**Flight Controller and Navigation** 1. Autonomous takeoff and landing 2. GPS-denied navigation stability 3. Autonomous return-to-base execution
Visual-Inertial Odometry (VIO) 1. Position and velocity estimation accuracy 2. Orientation tracking in dynamic flight conditions	**Integrated System Testing** 1. Coordination between sensing, estimation, and control subsystems 2. End-to-end autonomous mission execution 3. Robustness under environmental disturbances (lighting changes, sensor noise)

<u>Test Plan Summary Table</u>

Test No.	Test Name	Description	Procedure	Expected Outcome
1	Visual Feature Detection	Verify the camera's ability to detect target features	Simulate seeded features in the Gazebo arena and evaluate detection confidence	Correct detection in >95% of trials
2	Image Capture	Verify the quality of	Capture images at different	Images suitable

	Validation	captured verification images	altitudes and viewpoints	for feature validation
3	IMU Stability	Test IMU behavior in static and dynamic flight	Log IMU data during hover and maneuvers	Stable acceleration and rate estimates
4	VIO Position Tracking	Validate position and velocity estimation	Compare the VIO output with the commanded motion	Position error within acceptable bounds
5	Autonomous Takeoff	Verify safe autonomous takeoff	Execute takeoff sequence in simulation	Smooth ascent and stable hover
6	Autonomous Landing	Validate safe landing at base station	Perform repeated landing trials	Landing within the tolerance region
7	End-to-End Mission	Verify the full ASCEND mission	Execute takeoff → search → return → land	Mission completed without intervention
8	Resilience Testing	Test behavior under faults	Simulate sensor dropouts and low energy	Safe recovery and controlled landing

6. Brief of system specification

Sl. No.	Description	Specification
1.	Overall mass	1689.24
2.	Overall dimensions (mm)	500*500*206
3.	Power requirements(w)	243.16
4.	Flight time per charge	9.64 min
5.	T\W ratio	2.5
6.	Number of propellers and their size	4(10x45)

7. Overview of the emergency response system.

Emergency situation	Description	Solution
Power failure	An unexpected mid-flight power loss may cause the drone to crash, risking damage to both the drone and nearby areas.	Implement a battery health monitoring system and use an auxiliary battery to initiate an emergency landing before complete power loss.

Communication loss	i) A disruption in communication between the drone and the ground station can result in a potential loss of control. ii) Loss of communication between the Pixhawk and Jetson can disrupt autonomous flight operations.	i) Enable an autonomous return-to-home (RTH) function. Use pre-programmed safe zones as fallback destinations. ii) This issue is mitigated by utilizing PX4's offboard mode to maintain autonomous control during communication interruptions.
Obstacle collision	Inability to detect or avoid obstacles during flight may result in damage to the drone or surrounding objects.	Deploy real-time obstacle detection and avoidance algorithms using RANSAC-based point cloud analysis for precise avoidance.
Landing Zone Obstruction	Insufficient precision during landing can lead to improper alignment with the dock, causing obstruction or unsafe landing conditions.	Landing accuracy issues are mitigated using vision-assisted precision landing.
Environmental factors	Adverse conditions like water-ice, dry-ice, and global dust storms affect drone stability and camera vision.	Upon detection of camera degradation due to dust or environmental factors, the system activates an autonomous RTH sequence to restore sensor reliability.
System malfunctions	Failures in key components like theIMU, camera, or point cloud mapping, leading to operational errors.	Failures in essential components are managed through continuous health monitoring and fail-safe mechanisms that trigger autonomous RTH or controlled landing.

8. Project management

To ensure the successful execution of the ASCEND mission, the project is managed by assigning clear roles and timelines. The following management strategy has been established.

8.1 Team Roles and Responsibilities

The team is divided into functional roles, with each team member having a task assigned to them.

No.	Task	Main Responsibility	Deadline for Completion	Secondary Responsibility
1	Frame Assembly & Wiring	Poojith	February	Tushar
2	Hardware and software Integration	Tanmay	March	Ranjith
3	Simulation and Testing	Sanjana	February	Poojith
4	Docking station	Saalim	March	Tushar
5	ROS Implementation	Ranjith	February	Tanmay
6	Navigation and Mapping	Dhanushree	March	Neil
7	Detection and Validation	Laksh	April	Aman
8	Documentation and Report	Aman	A	Neil

8.2 Project Schedule & Milestones

The ASCEND project follows a milestone-driven schedule to ensure the timely completion of design, development, and validation activities in alignment with the IRoC-U 2026 competition timeline. A combined hardware–software Gantt-based planning approach is adopted to track progress and manage dependencies between subsystems.

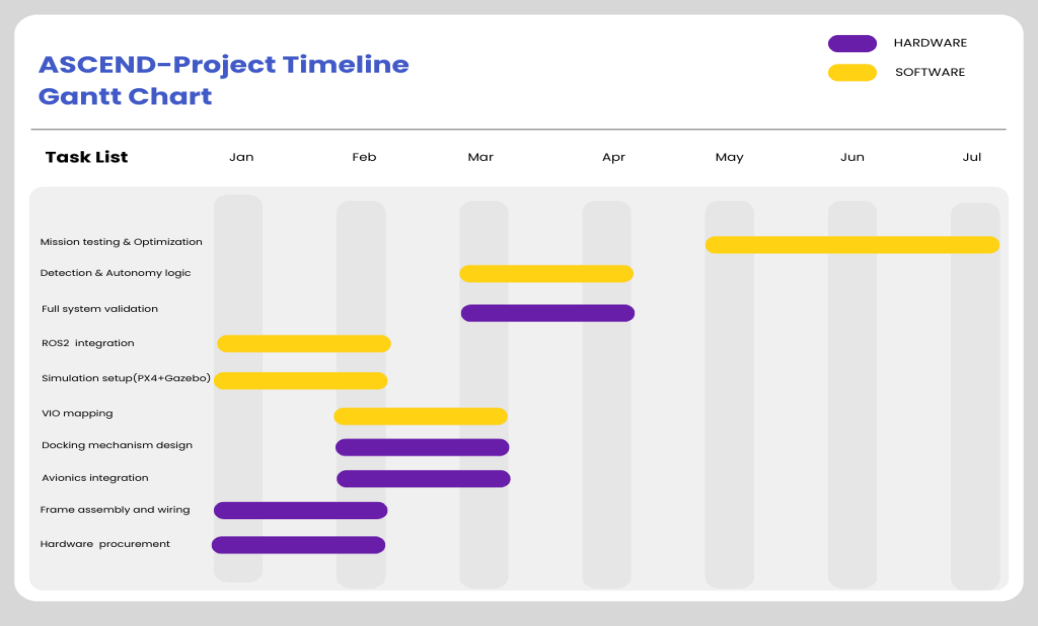

Fig 8.1: Ascend project timeline Gantt chart

The development is organized into four phases aligned with the IRoC-U 2026 timeline.

Phase I (Dec 2025–Jan 2026) focuses on system architecture finalization and high-fidelity simulation using PX4 SITL, ROS 2, and Gazebo to validate autonomy, perception, and navigation in a GPS-denied environment.

Phase II (Jan–Mar 2026) involves parallel hardware integration and software development, including airframe assembly, avionics integration, docking station development, and implementation of VIO, mapping, and feature detection.

Phase III (Mar–Apr 2026) conducts subsystem and full-system testing to validate autonomous takeoff, search, return, landing, data transfer, and charging under competition-like conditions.

Phase IV (Apr–Jul 2026) emphasizes performance optimization, robustness testing, and iterative refinement, culminating in a fully autonomous and competition-ready ASCEND system.

9. Novelty in the overall proposal

1. Optimized Drone Frame Design for Docking Compatibility

The quadrotor frame is intentionally designed to be lightweight and compact, ensuring efficient placement of all components while maintaining a well-balanced center of gravity. This streamlined layout enhances overall flight stability, and the landing structure is shaped to interface smoothly with the circular docking station, enabling precise and reliable autonomous docking and charging.

2. Charging Plate Cleaning and Drone Alignment Mechanism

The drone uses the geometric shape of the docking base and the metallic charging contacts as passive references for accurate landing, which are integrated into a Kalman filter–based state estimation approach. This allows precise landing without relying on high-precision localization systems. As the drone approaches the docking station, servo-controlled brush arms clean the copper charging plates to ensure reliable electrical contact. After landing, two brush arms and two alignment arms rotate to correct small positional errors, aligning the drone accurately over the charging interface. This approach improves docking reliability and repeatability.

3. Sensor fusion

A key novelty of the proposed ASCEND system is the sensor-fusion-based autonomy, which combines 1D LiDAR, Intel RealSense T265, and an onboard RPi 3 camera to achieve accurate operation in a GNSS-denied environment. The 1D LiDAR provides reliable altitude estimation during approach and landing, while the T265 is used for mapping and SLAM, enabling robust local pose estimation. The Raspberry Pi 3 camera is used for search operations.